

PYTHON SET-1

SCENARIO BASED QUESTIONS

1. Scenario: A system checks if a user is eligible to vote based on their age.

Logic:

1. Ask the user to enter their age.
2. Check whether the age is 18 or greater.
3. If the age is 18 or above, the person is eligible to vote.
4. Otherwise, the person is not eligible to vote.

2. Scenario: A program processes a list of numbers and needs to find the largest value.

Logic:

1. Take a list of numbers as input.
2. Assume the first number is the largest.
3. Compare it with each remaining number.
4. If a number is greater than the current largest number, make it the new largest.
5. Continue until all numbers have been checked.
6. Return the largest number.

3. Scenario: A company provides employees with a 10% bonus if their salary exceeds \$50,000.

Logic:

1. Ask for the employee's salary.
2. Check whether the salary is greater than \$50,000.
3. If it is greater than \$50,000, calculate a 10% bonus.
4. If it is \$50,000 or less, the bonus is zero.
5. Return the bonus amount.

4. Scenario: A program evaluates a number to determine if it is even or odd.

Logic:

1. Ask the user to enter a number.
2. Divide the number by 2 and check the remainder.
3. If the remainder is 0, the number is even.
4. Otherwise, the number is odd.

5. Scenario: A text-processing tool reverses a given word or sentence for formatting purposes.

Logic:

1. Ask the user to enter a word or sentence.
2. Start from the last character.
3. Read each character moving toward the first character.
4. Add each character to a new result.
5. When all characters have been processed, return the reversed text.

6. Scenario: A grading system determines whether a student has passed or failed based on their score.

Logic:

1. Ask the student to enter their marks.
2. Check whether the marks are 40 or greater.
3. If the marks are at least 40, the student has passed.
4. If the marks are below 40, the student has failed.

7. Scenario: A retail store offers a 20% discount if a customer's total order exceeds \$100.

Logic:

1. Ask for the customer's total order amount.
2. Check whether the order amount is greater than \$100.
3. If it is greater than \$100, calculate a 20% discount.
4. Subtract the discount from the original order amount.
5. If the order is \$100 or less, apply no discount.
6. Return the final amount to be paid.

8. Scenario: A banking system processes withdrawal requests and ensures the user has enough balance.

Logic:

1. Ask for the user's current account balance.
2. Ask for the amount they want to withdraw.
3. Compare the withdrawal amount with the available balance.
4. If the withdrawal amount is less than or equal to the balance, allow the withdrawal.
5. Subtract the withdrawal amount from the balance.
6. Display the remaining balance.
7. If the withdrawal amount is greater than the balance, do not allow the withdrawal and indicate that there is insufficient balance.

9. Scenario: A calendar system verifies whether a given year is a leap year based on standard leap year rules.

Logic:

1. Ask the user to enter a year.
2. Check whether the year is divisible by 400.
3. If yes, it is a leap year.
4. Otherwise, check whether it is divisible by 100.
5. If yes, it is not a leap year.
6. Otherwise, check whether it is divisible by 4.
7. If yes, it is a leap year.
8. Otherwise, it is not a leap year.

10. Scenario: A program filters out only even numbers from a given list.

Logic:

1. Take a list of numbers as input.
2. Create an empty result list.
3. Check each number in the given list.
4. Determine whether the number is divisible by 2.
5. If the remainder is 0, add the number to the result list.
6. Ignore odd numbers.
7. After checking all numbers, return the list containing only the even numbers.